ROWLAND AWADAGIN HERBERT-FAULKNER, PH.D

daginfaulkner.com | dagin_faulkner@berkeley.edu

PROFESSIONAL SUMMARY

Technology governance and implementation strategy professional with experience managing cross-organizational research teams, coordinating multi-stakeholder infrastructure initiatives, and advising public-sector decision-makers.

EXPERIENCE

University of California, Berkeley — **Berkeley, CA**
Researcher — 2018-2025

-Evaluated an equity-focused 20-mph residential-street intervention for San José DOT using large-scale Mobiliti simulation; estimated a 39% reduction in passthrough traffic on affected streets, while measuring spillover impacts on surrounding streets.
-Conducted regional-scale simulation research on Bay Area transportation network using Python, R, and geospatial methods; translated modeled traffic redistribution into policy-relevant measures of neighborhood traffic exposure and potential safety and equity benefits. Authored a policy brief and research report and presented findings to San José DOT.
-Studied Metropolitan Transportation Commission regional policy frameworks in relation to local transportation plans, examining how municipal traffic-management decisions intersect with regional objectives and network-wide effects.
-Collaborated with computer scientists, transportation researchers, and public-sector practitioners to develop decision-relevant approaches to transportation-network analysis and local agency capacity.
-Led multidisciplinary research involving 50+ government and industry stakeholder interviews examining AI, digital twins, smart-city technologies, and data governance, identifying technical, organizational, and governance requirements affecting implementation.
-Investigated municipal cybersecurity implementation, including technology procurement, third-party contracting, insurance, compliance, staffing, and institutional capacity; published findings through Berkeley's Center for Long-Term Cybersecurity.

DB Engineering and Consulting North America, Inc. — **San Francisco, CA**
Analyst, Technical Consulting — 2025-2026

-Developed agreements and implementation guidance for major California passenger-rail initiatives, defining responsibilities across public and private stakeholders and supporting interagency coordination, infrastructure planning, and phased service expansion.
-Coordinated directly with Caltrans staff to identify and resolve project-data discrepancies, reconcile infrastructure information, and refine agency-specific decision-support dashboards.
-Modeled the San Diego Trolley network and analyzed service and operating alternatives to inform medium- and long-term capital decisions involving fleet requirements, infrastructure needs, and viable service frequencies.

Just Cities Institute — **Berkeley, CA**
Project Manager (Fellow) — 2025

-Managed a seven-person cross-organizational research team comprising university researchers, undergraduate researchers, and a community partner; developed workplans, facilitated meetings and office hours, coordinated deliverables, and provided project updates to leadership for a CA Governor’s Office-funded initiative.
-Co-authored research on East Oakland traffic safety and mobility justice, examining community concerns and the equity implications of transportation investments; contributed to published analysis of the East Bay Bus Rapid Transit corridor.
-Led a 30-interview program evaluation commissioned by the California Air Resources Board, synthesizing perspectives from state-agency staff, academics, and advocacy organizations into findings on CARB Research Division programs.

-Conducted temporal and geospatial analysis of encampment sweeps and presented findings to the Mayor of Oakland and public- and community-sector stakeholders, translating analysis into evidence supporting advocacy for protections for unhoused populations.
-Developed and executed the scope of work for a comparative study of freeway-relocation projects focused on social and physical restoration in West Oakland.

Compass Education Group **Larkspur, CA**
SAT/ACT Tutor, Proctor, Administrative support 2022-Present

-Coordinates, plans, and conducts lessons with tutoring program managers, parents, and students
-Manages and administers in-person and remote standardized test sessions across multiple time zones

SELECTED PAPERS/PUBLICATIONS

Policy Brief: "Mobiliti—A New Tool to Guide Safer, More Equitable Traffic Management Strategies", UC Berkeley Institute of Transportation Studies, co-authored with Jane Macfarlane, PhD., Karen Trapenberg Frick, PhD., and Joan L. Walker, PhD. September 2025. https://bit.ly/4bWm8pb

White Paper: "The Transaction Costs of Municipal Cyber Risk Management", UC Berkeley Center for Long-Term Cybersecurity, published April 2024. https://bit.ly/4c3iFV6

SELECTED PRESENTATIONS AND MEDIA COVERAGE

2024 Intelligent Transportation Society of California (ITSCA) Annual Meeting: "Data-Driven Perspectives: An Exploration of the Impact of Local Transportation Management Strategies on Regional Traffic Dynamics"

"As Smart Cities Expand, So Do the Threats", by Robert Lemos for Dark Reading, February 6, 2024.
https://www.darkreading.com/ics-ot-security/smart-city-ecosystems-face-cybersecurity-shortfall-as-urban-populations-grow

EDUCATION

University of California, Berkeley **Berkeley, CA**
Ph.D. in City and Regional Planning 2025
Master of City Planning 2018
M.S. in Transportation Engineering 2017

Massachusetts Institute of Technology **Cambridge, MA**
B.S. in Civil Engineering; concentration in Urban Studies and Planning 2016